

Lesson 3 Ros Software Package

Creating information and service, writing publishers and subscribers and writing server and client is all based on Ros software package. In this lesson, we are going to learn how to constitute and create Ros software package.

In general, there are two ways to create software package, namely `roscatkin create pkg` and `catkin create pkg`. And we will teach you how to operate with `catkin create pkg`.

1. Components of Ros software package

Package.xml file meeting the Catkin specification and **CMakeLists.txt** file of catkin version must be included in catkin software package.

Package.xml offers meta-information related to this software package, such as its name, version, author, maintainer, etc. **CMakeLists.txt** is the input to **CMake** build system.

Note:

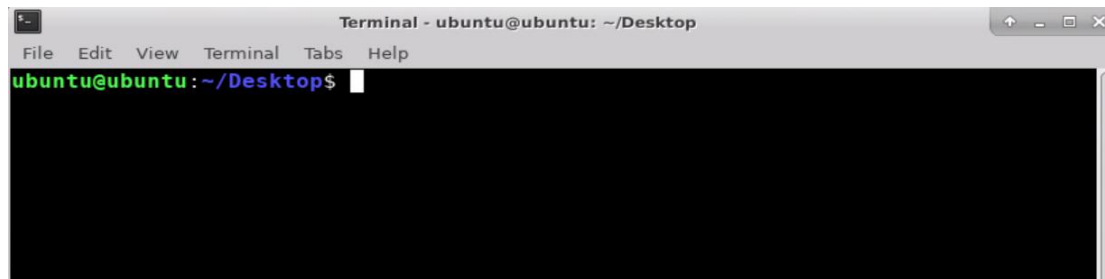
- If the newly created package belongs to meta-package, a boilerplate of **CMakeList.txt** is needed.

- Each package must have its own directory. That's to say, in the same directory, we cannot foist other packages to a package. Or place multiple packages in the same directory. A simple software package looks like this.

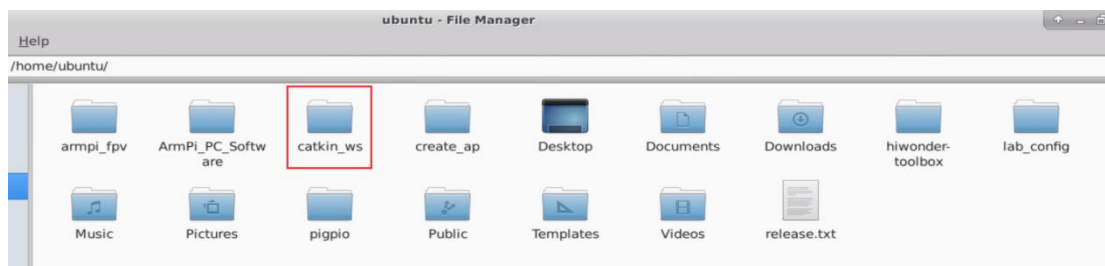
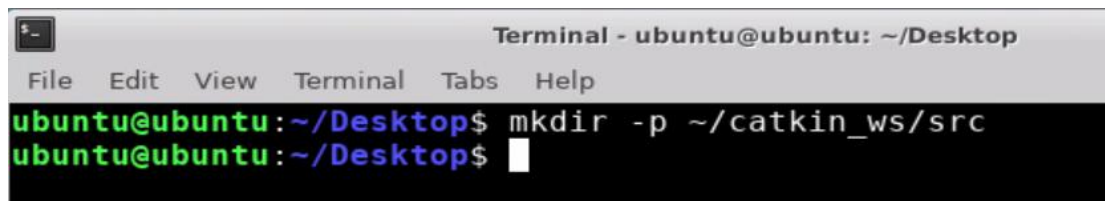
```
my_package/  
  CMakeLists.txt  
  package.xml
```

2. Creating Ros software package

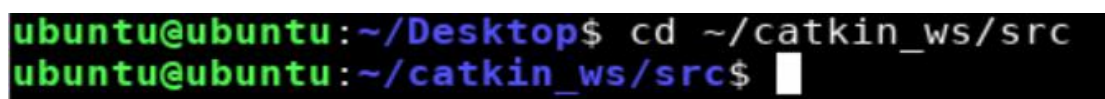
1) Before creating Ros software package, we need to build a workspace for catkin. Right-click on the remote desktop of NoMachine and select “**Open Terminal Here**” to open Linux terminal.



2) Enter “**mkdir -p ~/catkin_ws/src**” command in the opened terminal interface, then press “Enter”. Create a workspace called **catkin_ws** in this directory.



3) Enter “**cd ~/catkin_ws/src**” command and press “Enter” to get into the source file space, src, directory for catkin_ws.



4) Enter “**catkin_create_pkg beginner_hiwonder std_msgs rospy roscpp**” command and press “Enter”. Then create software package named

beginner_hiwonder and add dependencies, namely msgs, rospy and roscpp.

```
ubuntu@ubuntu:~/catkin_ws/src$ catkin_create_pkg beginner_hiwonder std_msgs
rospy roscpp
Created file beginner_hiwonder/package.xml
Created file beginner_hiwonder/CMakeLists.txt
Created folder beginner_hiwonder/include/beginner_hiwonder
Created folder beginner_hiwonder/src
Successfully created files in /home/ubuntu/catkin_ws/src/beginner_hiwonder.
Please adjust the values in package.xml.
ubuntu@ubuntu:~/catkin_ws/src$
```

- 5) Enter “cd..” command and press “Enter” to return to the directory above, ~/catkin_ws/.

```
ubuntu@ubuntu:~/catkin_ws/src$ cd ~/catkin_ws/src
ubuntu@ubuntu:~/catkin_ws/src$ cd ..
```

- 6) Enter “catkin_make” and press “Enter” to complete the software package creation in catkin workspace.

```
ubuntu@ubuntu:~/catkin_ws$ catkin_make
Base path: /home/ubuntu/catkin_ws
Source space: /home/ubuntu/catkin_ws/src
Build space: /home/ubuntu/catkin_ws/build
Devel space: /home/ubuntu/catkin_ws/devel
Install space: /home/ubuntu/catkin_ws/install
Creating symlink "/home/ubuntu/catkin_ws/src/CMakeLists.txt" pointing to "/opt/ros/melodic/share/catkin/cmake/toplevel.cmake"
####
### Running command: "cmake /home/ubuntu/catkin_ws/src -DCATKIN_DEVEL_PREFIX=/home/ubuntu/catkin_ws/devel -DCMAKE_INSTALL_PREFIX=/home/ubuntu/catkin_ws/install -G Unix Makefiles" in "/home/ubuntu/catkin_ws/build"
###
```

- 7) Enter “. ~/catkin_ws/devel/setup.bash” command and press “Enter”. Make configuration file take effect and add **catkin_ws** workspace to Ros environment.

```
ubuntu@ubuntu:~/catkin_ws$ . ~/catkin_ws/devel/setup.bash
ubuntu@ubuntu:~/catkin_ws$
```